

Table 5: Summary of main notations definitions.

Notation	Description
t	Diffusion time steps $t \in \{1, 2, \dots, T\}$
e_θ	Neural network parameterized with θ
$x \in \mathbb{R}^d$	Data point in d -dimensional space
f	predefined downsampling factor
R	Total number of communication rounds
r	Communication round $r \in [R]$
K	Total number of clients in the FL system
$D_k = \{x_i\}_{i=1}^{N_k}$	Local dataset of client $k \in [K]$ with a size N_k
$D \triangleq \bigcup_{k=1}^K D_k$	Global dataset with K clients
σ	Noise scale to control Gaussian noise injection
C	Clipping threshold to limit the norm of gradients
M	The number of samples for noise multiplicity

A Notation and Definition

This section provides a list of key notations in Table 5.

B Privacy Analysis

To calculate the privacy budget consumption of Algorithm 2, we employ the Gaussian Differential Privacy (GDP) [9] for theoretical analysis. We first provide the following lemmas, which are important primitives for the privacy analysis.

LEMMA 1 (ROBUSTNESS TO POST-PROCESSING [13]). *Let $\mathcal{M} : \mathcal{D} \rightarrow \mathbb{R}$ be a randomized mechanism that satisfies (ϵ, δ) -DP. If $F : \mathbb{R} \rightarrow \mathbb{R}'$ is an arbitrary deterministic or randomized mapping, then $F(\mathcal{M})$ also satisfies (ϵ, δ) -DP.*

LEMMA 2 (SUB-SAMPLING ON GDP [3]). *Let a randomized mechanism $\mathcal{M}$ satisfy μ -GDP, and $\mathcal{M} \circ \text{PoissonSample}_p$ denote the privacy amplification by Poisson sub-sampling with probability p . Suppose p is related to N such that $p\sqrt{N} \rightarrow v$, the N -fold composition of $\mathcal{M} \circ \text{PoissonSample}_p$ asymptotically satisfies $\hat{\mu}$ -GDP as $N \rightarrow \infty$, where*

$$\hat{\mu} = v\sqrt{\exp(\mu^2) - 1} = p\sqrt{N(\exp(\mu^2) - 1)}. \quad (15)$$

LEMMA 3 (COMPOSITION ON GDP [9]). *The n -fold composition of μ_i -GDP ($0 \leq i \leq n$) mechanism satisfy $\sqrt{\mu_1^2 + \dots + \mu_n^2}$ -GDP.*

LEMMA 4 (FROM GDP TO DP CONVERSION [9]). *A randomized mechanism $\mathcal{M}$ is μ -GDP if and only if $\mathcal{M}$ is (ϵ, δ) -DP for all $\epsilon \geq 0$, where*

$$\delta = \Phi\left(-\frac{\epsilon}{\mu} + \frac{\mu}{2}\right) - \exp(\epsilon)\Phi\left(-\frac{\epsilon}{\mu} + \frac{\mu}{2}\right). \quad (16)$$

LEMMA 5 (PARALLEL COMPOSITION OF DP [34]). *If there are n mechanisms $\mathcal{M}_1, \dots, \mathcal{M}_n$ computed on disjoint subsets whose privacy guarantees are $(\epsilon_1, \delta_1), \dots, (\epsilon_n, \delta_n)$ respectively, then any function of $\mathcal{M}_1, \dots, \mathcal{M}_n$ is $(\max_i \epsilon_i, \max_i \delta_i)$ differential private.*

In this project, to satisfy sample-level DP guarantee, each client adds Gaussian noise to their local data or model at each communication round. Given fixed DP-related parameters, clients incur identical privacy loss at each communication round. This subsection derives the cumulative privacy loss for each client across rounds. The results are summarized below.

THEOREM 5 (PRIVACY GUARANTEE OF PRIVFDM). *For any $\delta \in (0, 1)$, Algorithm 2 satisfies (ϵ, δ) sample-level DP with ϵ and δ satisfying:*

$$\delta = \Phi\left(-\frac{\epsilon}{\mu} + \frac{\mu}{2}\right) - \exp(\epsilon)\Phi\left(-\frac{\epsilon}{\mu} + \frac{\mu}{2}\right), \quad (17)$$

$$\mu = \sqrt{\mu_1^2 + \mu_2^2}, \quad (18)$$

with Phase I and Phase II satisfying μ_1 -GDP and μ_2 -GDP, respectively.

Proof Sketch. We first apply Poisson sub-sampling to select several batches from the dataset to strengthen privacy guarantee. Then we adopt autoencoder to selected batches for dimension reduction, following by perturbing the input latent in Phase I and model gradient in Phase II at each communication round to ensure sample-level DP for each client. We first calculate the GDP privacy budget in Phase I and Phase II for a single client. Then we use the composition theorem of GDP (Lemma 3) to combine the privacy loss of two Phases. Following that, we convert the calculated GDP to (ϵ, δ) -DP (Lemma 4). Finally, based on the parallel composition property of DP (Lemma 5), each client trains on a locally disjoint dataset and the DP parameters are consistent across rounds, leading to the same privacy loss for each client.

Using Theorem 3 and Lemma 2, we can derive the DP guarantees for Phase I as follows:

LEMMA 6. *Consider the noise scale σ_1 , batch size B_1 , the size of the private dataset N_k of client k , and the number of iterations $\frac{R_1 \times N_k}{B_1}$ after R_1 communication rounds. The total GDP privacy loss of a single client k in Phase I is:*

$$\mu_1 = \frac{B_1}{N_k} \sqrt{\frac{R_1 \times N_k}{B_1} (\exp(4\tilde{d}/\sigma_1^2) - 1)}. \quad (19)$$

PROOF. Let $\mathcal{M}_1$ denote the data perturbation mechanism in Phase I such that $\mathcal{M}_1(z) = z + \mathcal{N}(0, \sigma_1^2 \mathbf{I})$. Following Theorem 3, we can derive that $\mathcal{M}_1$ satisfies $2\sqrt{\tilde{d}}/\sigma_1$ -GDP, where $\tilde{d}$ is the scaled dimensions of a latent z . Moreover, we use Poisson sub-sampling for privacy amplification. Following Lemma 2, given sub-sampling rate $p_1 = \frac{B_1}{N_k}$ and total iterations $N_1 = \frac{R_1 \times N_k}{B_1}$ under one local training round, Phase I satisfies $\frac{B_1}{N_k} \sqrt{\frac{R_1 \times N_k}{B_1} (\exp(4\tilde{d}/\sigma_1^2) - 1)}$ -GDP. $\square$

Using Theorem 1 and Lemma 2, we can derive the DP guarantees for Phase II as follows:

LEMMA 7. *Consider the DP-SGD noise scale σ_2 , batch size B_2 , the size of the private dataset N_k of client k , and the number of iterations $\frac{R_2 \times N_k}{B_2}$ after R_2 communication rounds. The total GDP privacy loss of a single client k in Phase II is:*

$$\mu_2 = \frac{B_2}{N_k} \sqrt{\frac{R_2 \times N_k}{B_2} (\exp(1/\sigma_2^2) - 1)}. \quad (20)$$

PROOF. Let $\mathcal{M}_2$ denote the model perturbation mechanism (DP-SGD, Equation 2) in Phase II such that $\mathcal{M}_2(D) = f(D) + \mathcal{N}(0, C^2 \sigma_2^2)$, where C is the clipping threshold to bound the per-sample gradient sensitivity. Following Theorem 1, we can derive that $C^2 \sigma_2^2 = C^2/\mu^2$, so the $\mathcal{M}_2$ satisfies $1/\sigma_2$ -GDP. Moreover, we use Poisson sub-sampling for privacy amplification. Following Lemma 2, given sub-sampling rate $p_2 = \frac{B_2}{N_k}$ and total iterations $N_2 = \frac{R_2 \times N_k}{B_2}$ under one local training round, Phase II satisfies $\frac{B_2}{N_k} \sqrt{\frac{R_2 \times N_k}{B_2} (\exp(1/\sigma_2^2) - 1)}$ -GDP. $\square$

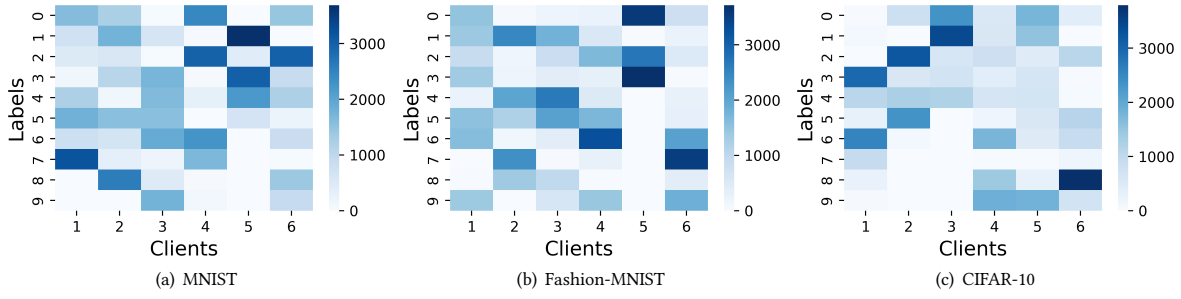


Figure 6: Label distributions under Dirichlet partitioning ($\alpha = 0.5$) for all clients in three investigated image datasets.

Table 6: The statistics of datasets used in the evaluation.

Dataset	Dirichlet α (Non-IID)	Clients	Training data (per client)	Dimensions	Category
MNIST	0.5	6	10000	$32 \times 32 \times 1$	10
Fashion-MNIST	0.5	6	10000	$32 \times 32 \times 1$	10
CIFAR-10	0.5	6	≈ 8333	$32 \times 32 \times 3$	10

Next, the post-processing property of DP (Lemma 1) ensures that any post-processing computation (e.g., aggregation at the server) on local updated model is differentially private as long as the training satisfies μ_1 -GDP and μ_2 -GDP. Then with Lemma 1 and composition theorem of GDP (Lemma 3), we can compute the overall privacy loss of a single client.

LEMMA 8 (PRIVACY LOSS OF THE CLIENT). *Consider a single client in Algorithm 2 with Phase I satisfies μ_1 -GDP and Phase II satisfies μ_2 -GDP, given $\delta \in (0, 1)$, a single client satisfies overall (ϵ, δ) -DP, where*

$$\delta = \Phi\left(-\frac{\epsilon}{\mu} + \frac{\mu}{2}\right) - \exp(\epsilon)\Phi\left(-\frac{\epsilon}{\mu} + \frac{\mu}{2}\right), \quad (21)$$

$$\mu = \sqrt{\mu_1^2 + \mu_2^2}, \quad (22)$$

where μ_1 is defined in Equation 19 and μ_2 is defined in Equation 20.

PROOF. Algorithm 2 contains two training phase for a single client. Following the composition theorem of GDP (Lemma 3), we can derive the overall GDP privacy loss μ of a single client $\mu = \sqrt{\mu_1^2 + \mu_2^2}$. Then, following the conversion theorem from GDP to (ϵ, δ) -DP (Lemma 4), we can compute the overall (ϵ, δ) -DP privacy loss for a single client. $\square$

In what follows, we give the proof of Theorem 5.

PROOF OF THEOREM 5. By Lemma 6, Lemma 7, and Lemma 8, the privacy losses incurred in Phase I and Phase II are each bounded. Since each client is allocated the same privacy parameter, applying the parallel composition property of DP (Lemma 5) and combining the results via Lemma 8, we conclude that the overall mechanism satisfies the claimed DP guarantee. This completes the proof. $\square$

Finally, since the privacy of the parameters of DMs is bounded by DP, as each client employs a differentially private DM to generate synthetic data, the resulting synthetic dataset satisfies (ϵ, δ) -DP by the post-processing property of DP.

C Implementation Details

This section provides detailed information on the label distribution, baselines, evaluation metrics, and configurations.

C.1 Details of Datasets

In this work, we use three public datasets, including MNIST [24], Fashion-MNIST [47], CIFAR-10 [23]. MNIST and Fashion-MNIST each consist of 70,000 grayscale images of handwritten digits and fashion products, respectively, with a resolution of $28 \times 28 \times 1$ pixels. To ensure compatibility with rescaling in autoencoder, all images are resized to $32 \times 32 \times 1$ pixels. Each dataset is split into 60,000 and 10,000 for training and testing. CIFAR-10 contains 60,000 colored images at a resolution of $32 \times 32 \times 3$ pixels, drawn from 10 object categories. The dataset is split into 50,000 and 10,000 for training and testing. To simulate data heterogeneity across clients, we adopt a Dirichlet distribution-based partitioning strategy. Specifically, the training data of each dataset is sampled for 6 clients as in prior works [5, 48–50] using the Dirichlet distribution ($\alpha = 0.5$) to simulate label distribution skew, while ensuring that each client has an equal number of training samples. The details of three datasets are presented in Table 6. Besides, Figure 6 illustrates the label distribution across clients under the Dirichlet-based non-IID partitioning scheme ($\alpha = 0.5$). Each heatmap visualizes how data samples from different classes are assigned to individual clients, providing a clear reference for the degree of data heterogeneity in each dataset.

C.2 Details of Baselines

Since there is no work focusing on differentially private federated diffusion models for image synthesis, this work compares with existing DP-FL methods and adapts state-of-the-art DP-DMs methods to the federated context. These methods do not rely on any external resources. Specifically, we evaluate our PrivFDM againsts four baselines:

- **DP-FedAvg [33]:** A traditional DP-FL algorithm that originally designed for client-level DP guarantee, we adjust it and directly apply DP-SGD to the local clients to ensure sample-level DP. We use the open-source codes of DPDMs [8] to implement DP-FedAvg. The hyper-parameters of training are the same as in DPDMs.

- **DP-FedProx:** FedProx [27] introduces an additional L_2 regularization term to local updating process to mitigate data heterogeneity among clients. We extend it with a sample-level DP guarantee by directly apply DP-SGD. We also use the platform of DPDMs [8] to implement DP-FedProx. All training hyper-parameters are the same as in DPDMs, except for the L_2 term is set to 0.1.
- **DPDMs (FL):** DPDMs [8] train diffusion models with DP-SGD. It reduces the gradient variance introduced by noise injection through a noise multiplicity strategy, which averages per-sample gradients computed over multiple noisy trajectories. We directly extend the official implementation¹ to the federated setting without modifying the original training hyper-parameters. The number of samples in noise multiplicity is set to 32 for MNIST and Fashion-MNIST, 8 for CIFAR-10.
- **DP-FETA (FL):** DP-FETA [25] adopts a two-stage training paradigm for diffusion models. In stage 1, limited mean representations of images perturbed with Gaussian noise are used to learn basic features. In stage 2, the model is fine-tuned on sensitive images using DP-SGD. We adapt the official implementation² to the federated setting, retaining all training hyper-parameters, with CIFAR-10 related parameters selected in prior work[16]. Besides, we observe that DP-FETA may leak label information during its stage 1. We modify this stage to perform unconditional training to avoid privacy leakage.

C.3 Details of Evaluation Metrics

- **Fréchet Inception Distance (FID).** FID quantifies the distance between the feature distributions extracted from real and generated images. Lower FID scores indicate higher visual quality and closer alignment to the real data distribution. We utilize the Inception V3 network³ provided by PyTorch to extract feature vectors from data.
- **Classification Accuracy (Acc).** To assess the utility of the synthetic data, we follow the prior works [8, 25, 46] to train a CNN classifier on the synthetic images for each client and evaluate on the real test data.

C.4 Details of Configurations

We implemented PrivFDM in PyTorch Lightning [36], building on the codebase by DP-LDMs [31]⁴ and LDMs[40]⁵. Specifically, we use the Opacus [51]⁶ to implement the DP-SGD algorithm. We conducted all experiments on 1 NVIDIA GeForce RTX 4090 GPU.

In the experiment, we build a neural network using a U-Net structure and train the network using the AdamW [32] optimizer. Moreover, we set diffusion time steps to $T = 1000$ and linearly assign forward noise schedule from 1.5×10^{-3} to 1.55×10^{-2} for β_t . Local training round is set to 1.

¹<https://github.com/nv-tlabs/DPDM>

²<https://github.com/SunniLee/DP-FETA>

³https://github.com/mseitzer/pytorch-fid/releases/download/fid_weights/pt_inception-2015-12-05-6726825d.pth

⁴<https://github.com/ParkLabML/DP-LDM>

⁵<https://github.com/CompVis/latent-diffusion>

⁶<https://github.com/pytorch/opacus>

Training. For MNIST and Fashion-MNIST, we set the downsampling factor $f = 8$ and transform each image into a $4 \times 4 \times 1$ latent space. In Phase I, we pre-train the network with a learning rate of $\eta_1 = 3 \times 10^{-4}$ and $R_1 = 3$ communication rounds at a batch size of $B_1 = 32$. In Phase II, we fine-tune the network with a learning rate of $\eta_2 = 6 \times 10^{-4}$ and $R_2 = 80$ communication rounds at a batch size of $B_2 = 4096$. Moreover, we set noise multiplicity $M = 32$, privacy failure probability $\delta = 1 \times 10^{-5}$ and clipping threshold $C = 1$.

For CIFAR-10, we set the downsampling factor to $f = 4$ and transform each image into an $8 \times 8 \times 3$ latent space. In Phase I, we pre-train the network with a learning rate of $\eta_1 = 3 \times 10^{-4}$ and $R_1 = 3$ communication rounds at a batch size of $B_1 = 32$. In Phase II, we fine-tune the network with a learning rate of $\eta_2 = 3 \times 10^{-4}$ and $R_2 = 30$ communication rounds at a batch size of $B_2 = 2048$. Moreover, we set noise multiplicity $M = 8$, privacy failure probability $\delta = 1 \times 10^{-5}$ while fixing clipping threshold $C = 1$. For the noise scales σ_1 and σ_2 used in training Phase I and Phase II under different privacy budgets, we present in Table 7.

Evaluation. To evaluate the fidelity of the synthetic image, we generate 20k images for each client to compute FID, and report the mean FID across all clients as the final evaluation. In the sampling procedure, following prior works [31, 40, 44, 46], we employ the DDIM sampler [42] and consider stochastic generation with sampling time steps $T' = 200$ for all sampling processes, including baselines. For the utility, we train a CNN classifier on the 20k synthetic images for each client and evaluate on the real test data. We report the average test accuracy across all clients as the final accuracy. Specifically, we train all classifiers using Adam optimizer with 50 rounds, batch size 128 and a learning rate of 3×10^{-4} .

Compared to centralized settings where 60k synthetic images are usually generated [8, 25, 46], we generate 20k samples for each client, as it is enough to (i) compute a reliable FID score and (ii) train an effective classifier with higher computational efficiency, enabling an effective and efficient comparison with baselines in the federated setting.

We list all the detailed hyperparameter settings in Table 8 and Table 9 for the autoencoder and diffusion models.

C.5 Details of Noise Multiplicity

In Section 4.3, we implement a noise multiplicity mechanism [8]. Specifically, we set noise multiplicity $M = 32$ for MNIST and Fashion-MNIST, and $M = 8$ for the more complex CIFAR-10 dataset. We also set $M = 8$ on CIFAR-10 for two baselines that applied this mechanism, DPDMs (FL) and DP-FETA (FL). This adjustment is due to the computational cost increasing linearly with M , since the denoiser needs to run M times, a factor not accounted for in the baselines. Following prior work [44], we adopt $M = 8$ on CIFAR-10 to achieve a better trade-off between generative performance and computational efficiency in federated settings.

C.6 Details of Model Sampling

We adopt the DDIM sampler [42] and perform stochastic generation with a fixed number of sampling steps $T' = 200$ for all sampling processes, including baselines. Note that two baseline methods, DPDM (FL) and DP-FETA (FL), originally employed DDIM or more advanced samplers with $T' = 1000$ steps. To ensure a fair technical

Table 10: Running time of DPDMs (FL), DP-FETA (FL), and our PrivFDM on three investigated image datasets.

Dataset	Module	DPDMs (FL)	DP-FETA (FL)	PrivFDM
MNIST	Pre-train	-	7.3h	5.8h
	Fine-tune	54.0h	39.0h	29.4h
Fashion-MNIST	Pre-train	-	7.3h	5.8h
	Fine-tune	54.0h	39.0h	30.5h
CIFAR-10	Pre-train	-	7.0h	4.0h
	Fine-tune	25.2h	10.0h	59.0h

Table 7: Combinations of noise scales in PrivFDM across various privacy budgets for three investigated image datasets.

Privacy	MNIST		Fashion-MNIST		CIFAR-10	
	σ_1	σ_2	σ_1	σ_2	σ_1	σ_2
$\epsilon = 1$	7.13	23.67	7.95	22.03	27.00	11.00
$\epsilon = 3$	5.80	7.65	5.80	7.65	18.00	3.95
$\epsilon = 5$	4.46	5.15	4.70	5.00	15.00	2.65
$\epsilon = 8$	4.00	3.51	4.00	3.51	13.00	2.00
$\epsilon = 10$	3.48	3.62	3.40	4.20	13.00	1.59

Table 8: Hyper-parameter settings for the autoencoders.

Dataset	MNIST	Fashion-MNIST	CIFAR-10
Input size	32	32	32
Latent size	4	4	8
f	8	8	4
z-shape	$4 \times 4 \times 1$	$4 \times 4 \times 1$	$8 \times 8 \times 3$
Channels	128	128	128
Channel multiplier	[1, 2, 3, 5]	[1, 2, 3, 5]	[1, 2, 3]
Attention resolutions	[32, 16, 8]	[32, 16, 8]	[32, 16, 8]
Batch size	50	50	50
Rounds	100	100	100

Table 9: Hyper-parameter settings for the diffusion models.

Dataset	MNIST	Fashion-MNIST	CIFAR-10
Channels	64	64	128
Channel multiplier	[1, 2]	[1, 2]	[1, 2, 2, 4]
Attention resolutions	[1, 2]	[1, 2]	[1, 2, 4]
Number of res blocks	1	1	2
Number of head	2	2	8
Learning rate η_1	3×10^{-4}	3×10^{-4}	3×10^{-4}
Learning rate η_2	6×10^{-4}	6×10^{-4}	3×10^{-4}
Batch size B_1	32	32	32
Batch size B_2	4096	4096	2048
Global rounds R_1	3	3	3
Global rounds R_2	80	80	30
Cond stage key	class label	class label	class label
Conditioning key	adm	adm	adm
Number of classes	10	10	10
Trainable parameters	3.3M	3.3M	70.4M

comparison and achieve a better trade-off between utility and computational efficiency, we standardize the sampling configuration across all methods.

D Comparison of Shared Common Knowledge

As shown in Figure 7, the cross-attention method produces semantically inconsistent images on MNIST and Fashion-MNIST, indicating a higher sensitivity to heterogeneous local distributions. In contrast, our label embedding can provide more straightforward, practical guidance in this private setting, thereby alleviating the adverse impact of the distribution shift.

E Comparison of Time Cost.

This section investigates the time cost of various methods. Table 10 presents the runtime usage of DPDMs (FL), DP-FETA (FL), and our PrivFDM. Compared to these two SOTA DP image synthesis methods, PrivFDM achieves an average of 34% and 23% lower total time costs on MNIST and Fashion-MNIST, respectively. Although PrivFDM incurs a higher time cost on CIFAR-10, it still achieves substantial improvements under strict privacy constraints. Specifically, as shown in Table 1, when $\epsilon = 1$, PrivFDM yields an average of 50% lower FID and 163% higher Acc than DPDMs (FL) and DP-FETA (FL), highlighting its effectiveness in maintaining better generative performance.

F Related Works

This section provides a complement to the related works discussed in Section 6. As there is currently a lack of research on differentially private federated diffusion models for image synthesis, we focus our discussion on two relevant lines of work: (i) differentially private diffusion models, and (ii) federated diffusion models, both in the context of image synthesis.

Differentially Private Diffusion Models. For differentially private diffusion-based image synthesis, existing methods primarily rely on DP-SGD [1] to achieve privacy guarantees. DPDMs [8] demonstrate how to train diffusion models using DP-SGD and introduce a technique called noise multiplicity to mitigate the increased gradient variance caused by the random noise inherent to DP-SGD. While this approach improves the privacy-utility trade-off for DP image synthesis, it incurs a linearly increasing computational cost with the multiplicity factor, which limits its scalability and applicability in broader scenarios. DP-LDMs [31] and DP-LoRA [44] utilize latent diffusion models [40] pre-trained on external public datasets and apply DP-SGD to fine-tune them on sensitive data. Both methods improve the privacy-utility trade-off by updating only a small subset of model parameters, particularly within the attention modules. PrivImage [26] further investigates the potential mismatch between the semantic distributions of public pre-training data and private fine-tuning data. It demonstrates that pre-training effectiveness largely depends on the similarity between public and sensitive datasets, underscoring that semantically aligned public data is more suitable for pre-training. dp-promise [46] leverages the inherent Gaussian noise from the diffusion model’s forward process to satisfy the DP guarantee, thereby reducing the additional noise required during fine-tuning. Unfortunately, the diffusion model is trained to predict the inherent Gaussian noise injected at each step accurately. This precise noise estimation can enable an adversary to invert the forward process and recover the original input images, thereby undermining the intended privacy guarantee of the first

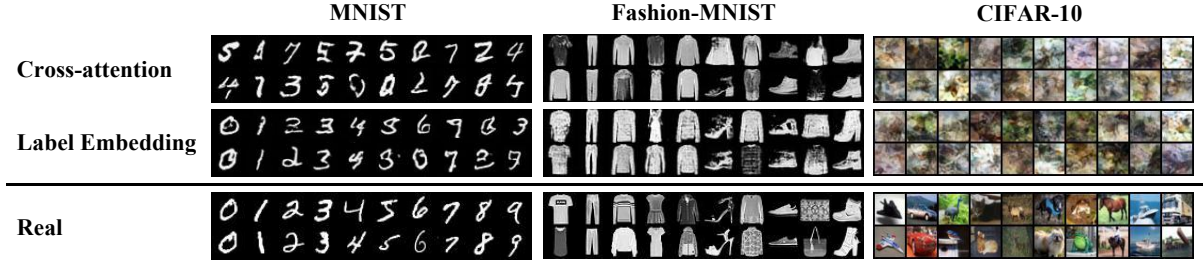


Figure 7: The synthetic images generated by cross-attention based label conditioning and label embedding based conditioning on MNIST, Fashion-MNIST, and CIFAR-10 under $\epsilon = 1$. The real image samples are presented in the last row for each dataset.

phase [16]. DP-FETA [25] employs a two-stage training framework. It perturbs the sensitive mean or mode images in stage one to learn basic features and improve the efficiency of DP-SGD in stage two. Nevertheless, DP-FETA suffers from significant privacy budget consumption, which limits training to a small number of mean or mode images during the first stage. When applied to federated clients, this limitation becomes more pronounced due to the reduced volume of local data, making the limited pre-training dataset overwhelmed by the increased noise.

Federated Diffusion Models. In the context of federated diffusion models, a main challenge lies in addressing data heterogeneity across clients. Recent works such as FedDISC [48] and FedDEO [49] explore this problem by leveraging pre-trained diffusion models in heterogeneous federated settings. These methods utilize pre-trained feature encoders (e.g., CLIP [38]) to extract features from input images or category labels. The extracted features are then aggregated via clustering, averaging, or training to form conditional representations that guide the generation process of diffusion models. The

goal is to synthesize client-specific datasets that better align with each client’s local distribution. FedLMG [50] adopts a classifier-guided generation approach, where classifiers trained locally on each client serve as intermediaries. A pre-trained diffusion model is conditioned on these classifiers to generate synthetic data that reflects client-specific semantics. FedBiP [5] addresses data heterogeneity by uploading only the client-specific latent representations and locally updated label embeddings to the server, which are then used as conditions to guide the generation process of a pre-trained latent diffusion model tailored to the client distributions.

However, despite their federated setting, these methods do not provide formal privacy guarantees for the local clients. Some even assess privacy risks based on visual dissimilarity between synthetic and real images, which can be misleading and give a false sense of privacy [52]. The lack of privacy guarantees limits their applicability in sensitive domains.